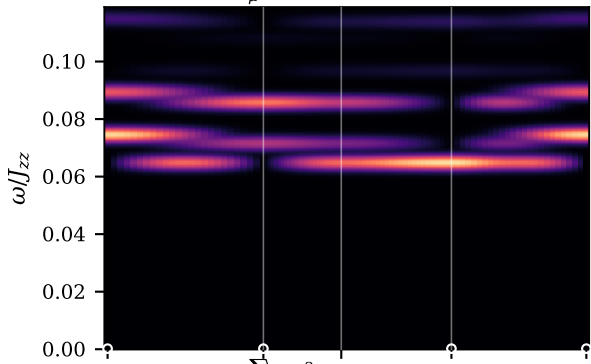
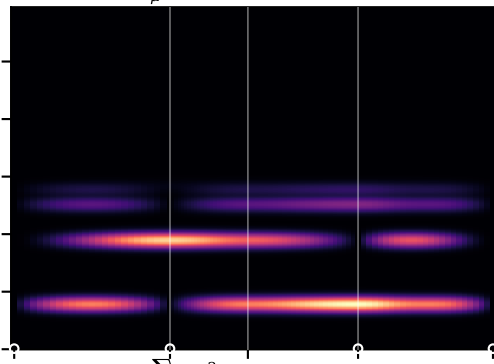


FCC-32 longitudinal response along a connected path, $J_{\pm}/J_{zz} = -0.200$, 144 states

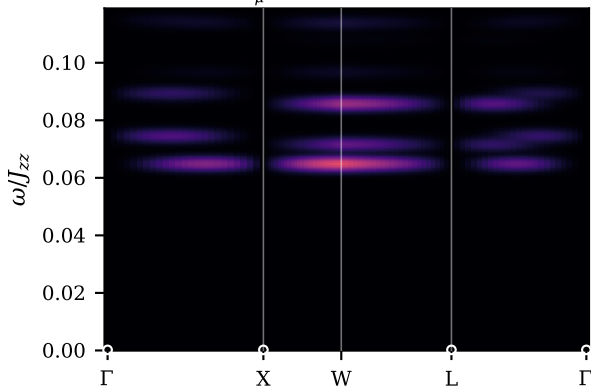
(a) periodic, $\sum_{\mu} |A^{\mu}|^2$



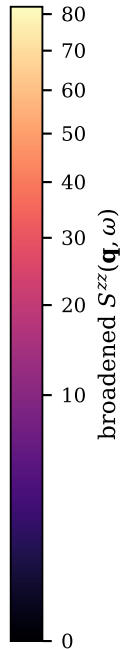
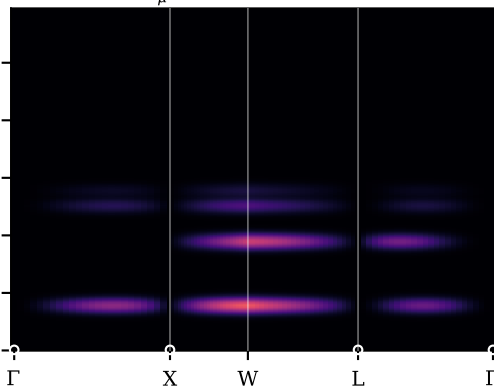
(b) masked, $\sum_{\mu} |A^{\mu}|^2$



(c) periodic, $|\sum_{\mu} A^{\mu}|^2$ (U(1)-protected)



(d) masked, $|\sum_{\mu} A^{\mu}|^2$ (U(1)-protected)



wave vector along Γ -X-W-L- Γ

wave vector along Γ -X-W-L- Γ